

	A	B	C	D	E	F	G	H	I	J
1	From the data of marks of students in test given below, test whether the average marks									
2	of a student is atleast 60 or not at 90% confidence level.									
3	55	65	60	62	63	45	70	75	70	65
4										
5	Solution:- Here, we have									
6	55	65	60	62	63	45	70	75	70	65
7		Sample size(n)=		10	=COUNT(A6:J6)					
8		Pop. Mean (μ)=		60						
9		Sample Mean($\bar{x}$)=		63	=AVERAGE(A6:J6)					
10		Sample SD(S) =		8.485	=STDEV.S(A6:J6)					
11		Here , we set up Hypothesis as								
12		$H_0 : \mu_1 > 60$ i.e the average marks of a student is atleast 60.								
13		$H_1 : \mu_2 \leq 60$ i.e the average marks of a student is more than 60.								
14		Under H_0 ,test statistics								
15			S.E($\bar{x}$) =	2.68	=D10/SQRT(D7)					
16			T_{cal} =	1.118	=(D9-D8)/D15					
17			C.L.($1-\alpha$) =	0.1						
18			(α) =	0.90	=1-D17					
19			d.f. =	9	=D7-1					
20			T_{tab} =	1.833	=T.INV.2T(D17,D19)					
21		Decision: - Since $T_{cal} < T_{tab}$,we reject H_1 and accept H_0 with the								
22		conclusion that the average marks of a student is more than 60.								
23										
24			Name :Hari Bashyal							